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Abstract

Accurately predicting football matches and identifying mispriced odds continue to be major challenges for both analysts and the betting sector. This study uses a five-season panel of 1,390 Belgian Jupiler Pro League fixtures (2019/20 to 2023/24), enhanced with 111 match variables, including rolling form, adaptive home advantage, Elo-style squad ratings, and k-means context clusters, to evaluate five standard classifiers: naïve Bayes, multinomial logistic regression, SVM, random forest, and XGBoost.

Following leakage control and random oversampling of draws and away wins, the models were optimized using nested cross-validation and assessed on an unutilized 2023/24 hold-out set ($n = 239$). The Random Forest algorithm showed superior discrimination (macro ROC AUC = 0.696; accuracy = 54.0%) and calibration (log loss = 0.986; Brier = 0.590); however, all algorithms encountered challenges with the draws (recall $\leq 25\%$). Feature importance analysis identified recent defensive strength (three match "shots against"), corner creation, and overall squad strength as predominant indicators. The comparison of model implied probabilities with closing odds from six bookmakers indicated average over rounds ranging from 1.9% (Max Bet) to 7.0% (Bet Victor). While accuracy alone does not ensure profitability, a flat stake backtest showed that naïve Bayes could identify undervalued draws into the highest cumulative return (€ 40.74 at MaxBet).

These findings show that carefully calibrated standard machine learning algorithms may assess market efficiency in a balanced league and reveal areas of bookmaker biases. Future endeavours should integrate player tracking and in-play data with sequence models to elevate performance over the current 50 to 55% threshold, while incorporating responsible gambling measures.

Key Words: Jupiler Pro League; football match outcome prediction; machine learning; sports betting markets; bookmaker bias and mispricing; random forest; probability calibration.

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List of Abbreviations

Table 1: List of Abbreviations

Abbreviation	Full Form
ML	Machine Learning
RF	Random Forest
SVM	Support Vector Machine
XGBoost	Extreme Gradient Boosting
NB	Naïve Bayes
LR	Logistic Regression
FTR	Full Time Result
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
TP	True Positive
FP	False Positive
FN	False Negative
OOB	Out-of-Bag
KPI	Key Performance Indicator
PS	Pinnacle Sports
VC	Bet Victor
B365	Bet365
BW	Bet and Win
Max	MaxBet
Avg	Bet Brain
EU	European Union
COVID-19	Coronavirus Disease 2019
CSV	Comma Separated Values

1 Introduction

Football (soccer) is one of the most popular sports in the world, passionately followed by billions of people worldwide. During the 2022 FIFA World Cup, for example, an estimated five billion people engaged with the content of the tournament, reflecting the immense global reach of the sport (FIFA, 2023). A key part of football’s appeal is its dynamic and uncertain nature, virtually anything can happen in a match. Upsets are common, and even historically dominant teams can suffer unexpected defeats. This competitive uncertainty has long been thought to fuel fan interest. According to Buraimo and Simmons, 2015, the ”uncertainty of outcome” hypothesis shows that football has flourished as a spectator sport because spectators find thrill in the unpredictable nature of results. In the 2014 World Cup semi-final, for instance, underdog Germany easily defeated Brazil 7-1, one of the most spectacular upsets in football history. Episodes like this show how hard it is to predict match results and why it is so hard to make precise predictions in football.

While football’s unpredictability is celebrated, it also presents a tempting challenge: Can match outcomes be predicted with any consistency? This question carries both theoretical and practical significance. For example, the foundation of the growing sports betting market is the attraction of prediction and the assumption of unpredictability. Bettors and bookmakers alike attempt to forecast results for financial gain, making outcome prediction a high stakes endeavor. The global football betting market has grown rapidly in recent years, with millions of fans trying to leverage their football knowledge to beat the odds (Constantinou et al., 2013). However, the inherent unpredictability of football means that even informed fans are often wrong more than right, and bookmakers typically retain an edge through careful odds setting (Štrumbelj and Šikonja, 2010). The Belgian Jupiler Pro League, in particular, offers a compelling case study to explore these issues. As Belgium’s top tier football league, the Jupiler Pro League features a competitive balance of teams and a rich history of surprise outcomes. Several different clubs have won the Belgian championship over the past decade, suggesting a league where matches are not dominated by a single ”powerhouse” and where underdog victories are plausible. This competitive parity, combined with the availability of detailed match data, makes the Jupiler Pro League an ideal context for developing and testing predictive models. By focusing on this league, the present study can investigate the predictability of the outcome in a real world setting that exemplifies the excitement and uncertainty of football.

2 Research Objectives

With the aforementioned background, this research pursues two primary objectives. Firstly, it aims to develop a robust machine learning model to predict match outcomes in the Belgian Jupiler Pro League and to evaluate its predictive accuracy against both baseline benchmarks and bookmaker odds. The focus will be on maximizing predictive performance through careful feature selection, model tuning, and validation, thereby assessing the extent to which football matches in this league can be accurately forecasted using historical data and statistical learning techniques. Second, the research seeks to investigate potential manipulations or biases in betting odds for the Jupiler Pro League. By comparing the model predictions (implied probabilities of home, draw, or away wins) and the odds offered by the bookmakers, the study will examine whether the odds accurately reflect the realities on the field or if there are consistent deviations. Such deviations could indicate the presence of betting market biases (for example, systematic under- or over-estimation of certain teams or outcomes) or even deliberate odds adjustments.

In particular, the analysis will explore whether there are instances in which the predictive model identifies value bets (that is, cases in which the bookmaker's odds appear mispriced relative to the probabilities of the model) and whether those instances can be linked to any external factors or irregularities. By meeting these objectives, the thesis aims to both improve predictive accuracy in a challenging sports domain and shed light on the reliability and fairness of the associated betting markets in order to increase public awareness of the risks of sports betting.

3 Background of the Study

3.1 Historical Approaches to Football Match Prediction

Rating systems and statistical models were the primary tools of early football match prediction research. One fundamental method was to use Poisson distributions to model scores. [Maher, 1982](#) developed a model with team specific offensive and defensive parameters that treated home and away goals as independent Poisson random variables. This independent Poisson framework provided a baseline for estimating the probabilities of match outcomes. [Dixon and Coles, 1997](#) later improved on the Maher model by accounting for the correlation between teams' scores, noting that the basic Poisson model tended to underestimate the frequency of low-scoring draws (for instance, 0 - 0 or 1 - 1) . Their bivariate Poisson formulation corrected for this issue, resulting in more accurate probabilities and even revealing minor inefficiencies when compared to bookmakers' odds. Another popular rating-based method is the Elo rating system (originally developed for chess), which assigns each team a strength rating that updates based on match results. Elo ratings have been applied to football to predict winners by comparing the difference in ratings between two teams. For example, during the 2014 World Cup, Brazil and Croatia had Elo Ratings of 2113 and 1787, respectively. Higher ratings indicate stronger teams, helping compare team strengths objectively ([Lyons, 2014](#)). This shows that Elo ratings are a substantial predictor of results; however, they are noticeably inferior to using bookmakers' odds as predictors, highlighting that while traditional ratings encapsulate past performance, they may miss other information that betting markets capture.

Markov models have also been explored to incorporate the sequence of play within a match. [Hirotsu et al., 2002](#) proposed a four-state Markov chain representation of a game, with states defined by which team has possession or scores a goal. Their model evaluated teams' offensive and defensive strengths (and home advantage) by analyzing transitions between these states during matches. Such state-space models could capture dynamic aspects of play (for instance, momentum shifts), offering a richer depiction of match progress than static models. However, these traditional statistical approaches all have notable limitations in capturing football's complexity. The Poisson-based models assume fixed team strengths and independent scoring events, which is a simplification of reality. In fact, one limitation is that many early models are static, treating the attack/defense parameters of the teams as constant over time ([Dixon and Coles, 1997](#)). In contrast, real team form can fluctuate during a season. They also struggle to represent

interactions between opposing teams; for example, treating the goals of each side as independent random variables ignores the inherent dependency of the game (if one team dominates possession, it is likely to suppress the scoring of the other). Rating systems like Elo condense performance into a single number and typically do not account for contextual factors such as line-up changes, injuries, or tactical nuances. Even the Markov chain approach is constrained by how states are defined (it might capture possession and goals, but not smaller events), and can become computationally complex with more detailed state definitions. In summary, while historical models like Poisson regression, Elo ratings, and simple Markov chains laid important groundwork for quantitative predictions in football, their simplifying assumptions limit their ability to capture the full tactical and interactive complexity of a 90-minute match. This recognition of limitations opened the door for more advanced, data-driven techniques to improve predictive performance.

3.2 Machine Learning in Football Outcome Prediction

In recent years, ML has emerged as a powerful tool in sports analytics, offering new ways to handle the problem of predicting the result of matches (Fu, 2023). Traditional statistical approaches to football Prediction (for example, Elo ratings and Poisson regression models) have achieved some success but often struggle to capture the complex, nonlinear interactions of variables that influence a game’s outcome (Kovalchik, 2020). Machine learning techniques such as tree-based models, ensemble methods, and neural networks can ingest large historical datasets and uncover subtle patterns or combinations of features that humans might overlook. In the football context, researchers have applied ML algorithms to model the outcomes of matches with promising modest results. For example, gradient boosted tree models and other advanced classifiers have achieved prediction accuracies in the range of approximately 52–54% for match results(home team win, draw, or away team win), outperforming naive baselines and approaching the accuracy of professional bookmakers (Hubáček et al., 2019; Holmes and McHale, 2024).

According to Hubáček et al., 2019, it showed that a carefully tuned ensemble model could correctly predict about 53 to 54% of match outcomes across several European leagues, slightly better than conventional rating systems. Similarly, Holmes and McHale, 2024 developed a player rating based predictive model that reached roughly 53% accuracy in forecasting English league matches, a level comparable to the odds-implied forecasts of betting markets. These studies show both the potential and limitations of current ML methods: potential, in that data-driven models can learn important predictors (such as team strength, recent form, or player performance

metrics) and sometimes even out-perform betting odds under specific conditions (da Costa et al., 2022); and limitations, in that even state-of-the-art models still hover around 50 to 55% accuracy in a sport where random events and unmodeled factors play a considerable role. Notably, recent work by Luiz et al., 2024 found that while football outcomes are difficult to anticipate, certain leagues and scenarios are more predictable than others, suggesting that clever feature engineering and league-specific insights can improve forecast reliability. Overall, the application of machine learning in football analytics holds great promise, not only for improving predictive accuracy but also for identifying which variables (team statistics, player ratings, tactical factors, etc.) most strongly influence match results. This knowledge can feed back into the sport, informing coaching decisions and talent evaluation.

3.3 Betting Market Efficiency and Biases

Beyond pure prediction, machine learning offers benefits in analyzing the betting market itself. If a predictive model can estimate match outcome probabilities with reasonable accuracy, these estimates can be compared against bookmakers' odds to identify potential inefficiencies or biases. Prior research indicates that while betting odds are generally well calibrated forecasts of match outcomes, anomalies do occur (Hubáček et al., 2019). For instance, there is evidence of systematic biases like the favorite longshot bias, where bettors tend to overvalue long shot outcomes, leading to distorted odds (Myrsten, 2021). Moreover, the integrity of sports competitions can sometimes be undermined by manipulation of outcomes for betting gain (match fixing), which often leaves tell tale patterns in odds movements. Machine learning methods have been employed to detect such irregular betting patterns, a form of predictive modeling aimed not at the match result itself, but at identifying when odds behave abnormally relative to expected probabilities (Kim et al., 2024). Also, Kim et al., 2024 developed an anomaly detection system that flags matches with suspicious odds fluctuations (which may indicate insider information or fix), achieving high classification accuracy in distinguishing normal vs. abnormal betting behavior. These applications highlight that modeling and prediction in football serve multiple stakeholders: not only can they predict sporting outcomes, but they can also enhance market transparency and integrity by spotlighting discrepancies between data-driven predictions and betting odds. Another prevalent bias involves home advantage and sentimental favorites. Bettors tend to overbet popular teams or home teams, sometimes independent of true quality. Sportsbooks are known to “shade” odds (adjust them away from fair value) to exploit these tendencies (Feddersen et al., 2016). For example, a

hometown team might have its odds lowered (payoff reduced) because the bookmaker anticipates heavy local betting interest regardless of value. Such practices mean odds are not set purely to balance books, but to maximize bookmaker profit by leveraging bettor bias ([Levitt, 2004](#)).

3.4 Societal Impacts of Sports Betting

Several moral questions and societal hazards are brought up by the growth of sports betting, which is partly driven by analytics and machine learning. The most common of these are addiction and problem gambling. Research shows that problem gambling is more common among sports bettors than among gamblers in general; one comprehensive assessment concluded that the prevalence of problem gambling was at least twice as high among sports bettors as it was among all gambling populations ([Winters and Derevensky, 2019](#)). According to the National Council of Problem Gambling, in the US, an estimated 1% of adults (around 2.5 million people) meet criteria for severe gambling disorder in a given year, with another 2–3% having milder gambling problems. The rise of online platforms and live betting (in play) aggravates these risks. In-play betting allows wagering on dozens of in-game events (next goal, corner kicks, etc.) with rapid resolution. This high frequency, instantaneous feedback loop can accelerate chasing losses and impulsive bets ([Winters and Derevensky, 2019](#)). Research confirms that in-play bettors tend to show greater gambling severity and harm than traditional bettors ([Vieira et al., 2023](#)).

Ethical concerns are also raised by concerns about fairness and openness in the odds-setting process. There may be a sense of unfair advantage because bettors frequently don't understand how bookies calculate odds or modify lines. In an effort to guarantee that bettors are aware of the actual house edge and odds, calls for increased openness in odds-setting procedures and advertising honesty have intensified. Furthermore, there is an obligation to maintain integrity because match-fixing and the exploitation of insider knowledge, which compromise fair play, are growing issues as betting volume rises. In order to preserve confidence in the sports and betting industries, stakeholders (leagues, sportsbooks, and data scientists) must work together ethically to identify and stop corruption, maintaining trust in both the sport and betting markets.

4 Materials and Methods

4.1 Data Collection, Features and Licensing

The data set used for this study includes match data from the Belgian Jupiler Pro League from 2019/2020 to 2023/2024 over five seasons. The main source of match related information, such as results, goals, other key events, and betting odds, was sourced from football-data.co.uk. This source provides football data free of charge for personal use, but users are encouraged to review their terms and conditions for any specific restrictions on usage ([Football-data.co.uk](https://football-data.co.uk), 2024).

Additional performance metrics for each team, including attack, midfield, defence, and overall ratings, were obtained from Sofifa.com (this was obtained using Power Query to source directly from the site). SoFIFA provides these metrics "as is" and does not guarantee their accuracy, so users should verify the data independently before relying on it for any purpose ([SoFIFA-Pro League FC 25](https://Sofifa.com)).

Furthermore, the league positions of each team before every match were collected from Transfermarkt.com. While Transfermarkt data is often shared under the Creative Commons Zero licence through third party repositories, users should consult Transfermarkt's official terms or seek permission for data extraction ([Transfermarkt-Pro League](https://Transfermarkt.com)).

4.2 Data Description and Cleaning

The combined data set initially contained 1,462 matches downloaded from FootballData.co and covered five Belgian Pro League seasons. The 2019/20 season was cut short by the COVID-19 outbreak, which featured only 232 matches. Subsequent structural changes expanded the league to 18 teams without play-offs, making a total of 306 games per season. The 2023/24 season reverted to 16 teams, making a total of 312 games per season due to the championship and relegation play-offs. The 72 play-off matches from 2023/24 were excluded to maintain a consistent competition format.

After this adjustment, the dataset comprises 1,390 matches, each described by 111 features and involving 22 clubs that appeared across the seasons. Variables include half-time and full-time goals for both teams, home and away team identifiers, and betting odds from ten bookmakers covering outcomes such as home win, draw, away win, and over/under 2.5 goals. Missing data analysis reveals near complete coverage for most fields. The main gaps occur in the odds supplied by Interwetten (5.8% missing) and William Hill (23%), These two bookmakers were left out.

4.3 Feature Engineering and Feature Selection

4.3.1 Feature Engineering

The match log was treated as a time series ordered by kick-off date. All dynamic features were generated with causal windows so that only information available up to the match t was used:

- 1 **Rolling average form** ($n = 3$): For each team, rolling means were calculated over the three most recent league matches. To capture each team’s short-term momentum, every match was first assigned a unique identifier, and home and away performances are merged into a single, chronologically ordered dataset. For each team, prior match statistics are shifted so that the current game is excluded, then averaged over the three most recent fixtures. This three match window smooths out one off spikes while still reflecting evolving form. The resulting home and away specific averages replaced raw counts, and redundant fields were removed, and the new form indicators were rejoined to the dataset, producing 28 new features of recent trends in goals scored, shots, fouls, cards, and corners.
- 2 **Team form** ($n \leq 5$): This feature captures a team’s momentum by averaging the points earned across its up to five most recent fixtures, whether played at home or away. Each win yields three points, a draw one point, and a loss zero points. The calculation adapts when fewer than five past matches exist, ensuring meaningful early season matches.

$$f = \min(5, \#\{\text{previous matches}\}), \quad \text{Form}_t = \frac{\sum_{i=1}^f P_{t-i}}{3f} \quad (1)$$

Where $P_{t-i} \in \{0, 1, 3\}$ is the point total from the i -th most recent match, and the index ranges from 0 (no points) to 1 (maximum points).

- 3 **Home advantage (adaptive $n \leq 5$)**: This metric quantifies a team’s recent performance on home ground by averaging the points earned in its up to five most recent home matches. For each prior fixture, three points are awarded for a win, one for a draw, and zero for a loss. If fewer than five home games have been played, the calculation scales to the number available, ensuring meaningful measurement from the season’s start.

$$m = \min(5, \#\{\text{previous home matches}\}), \quad \text{HA}_t = \frac{\sum_{i=1}^m P_{t-i}}{3m}, \quad (2)$$

where $P_{t-i} \in \{0, 1, 3\}$ represents points earned in the i -th most recent home match. The index ranges from 0 (no points) to 1 (maximum points).

4 K-means Clustering: K-means clustering is a widely used algorithm for partitioning a dataset into K distinct, non overlapping clusters based on similarity. To implement K-means clustering, the number of clusters K must first be specified (Hastie et al., 2023). The algorithm then assigns to assign each observation to one of the K clusters in a way that minimizes within-cluster variation, often calculated as the sum of squared Euclidean distances between observations and their respective cluster centroids. To perform K-means clustering on the data, a 21 feature matrix that described the immediate context of each match, including: Team form, Team league position, and three-match rolling tallies for goals, shots, shots-on-target, and corners for both teams. Every variable was z-scaled to prevent faster moving shot counts from controlling distance computations, which is a crucial safety measure for algorithms based on Euclidean principles. The games were continually moved to the closest centroid using the method, which also minimised the sum of squares within the cluster (inertia).

4.3.2 Feature Selection

Before modelling, the match outcome was set on a standard three point scale from the home team’s perspective: a home win ($H = 3$), a draw ($D = 1$), and an away win ($A = 0$). The resulting ordinal target (`Results_home`) properly leads into both regression and classification algorithms, keeping competitive meaning. To avoid target leakage, all columns that give information after the final whistle were removed. These included 57 post match statistics such as FTR, flags, corners (Home/Away), and cards (Home/Away). Retaining such variables would allow the model to ”peek” at the answer, producing overly optimistic results.

Also, to reduce redundancy in the remaining pre-match feature set, we computed Spearman rank correlations and removed or merged any pairs with $|\rho| > 0.8$. Spearman was chosen over Pearson because many engineered features (for example, rolling counts, ordinal encodings) are non-normally distributed and can relate to one another in monotonic but non-linear ways; the rank based measure is less subject to outliers and better reflects those relationships. Heatmaps of the characteristics (grouped eleven at a time beside the target variable) offered an interpretive discovery, indicating that multicollinearity had been addressed before the data entered the

modelling pipeline.

4.4 Data preprocessing

This section transforms raw match data into a clean, model-ready dataset by handling categorical labels, filtering out leakage, normalising continuous variables, and partitioning into training and testing sets. To begin, the two “Build-up Style” columns (home and away) and team cluster labels were converted into integer codes via label encoding, transforming qualitative patterns into reproducible numerical features. Columns that would leak future information, such as full-time scores, betting odds, and public percentages, were removed to focus on predictors available before each match.

Continuous data (for example, home advantage, team form, and rolling average form) were first imputed (mean filling) where missing, then standardised using z-scores (zero mean, unit variance) so that features with large numeric scales could not dominate model training. After preprocessing, the data was split chronologically at July 28, 2023: all matches before that date formed the training set (1,125 matches), and those on or after formed the test set (239 matches).

The original training set class counts were:

$$\text{Draw (0)} = 253, \quad \text{Home Win (1)} = 486, \quad \text{Away Win (2)} = 386.$$

Because draws and away wins (slightly lower) were underrepresented relative to home wins, random oversampling was applied to both Draw (0) and Away Win (2) classes so that each matched the Home Win (1) count of 486. Using Scikit Learn’s `RandomOverSampler`, draw and away win samples were duplicated with replacement until each minority class contained 486 instances. No resampling was performed on the test set, which retained its original distribution of 65 draws, 108 home wins, and 66 away wins. The resulting balanced training set consisted of 1,458 matches (486 per class). All hyperparameter tuning and model fitting were performed on this balanced training set, and final evaluation was conducted on the untouched test set of 239 matches.

4.5 Machine Learning Models

To predict whether a game will finish in a home victory, tie, or away victory, 5 machine learning models were built. To avoid overfitting, each model was adjusted using grid search combined with cross-validation. To find differences between the model’s assessment of a game and bookmaker odds, predicted class probabilities are provided by models such as logistic regression, random forest, and XGBoost. Furthermore, tree-based techniques are used to generate feature importance scores, allowing for a clear identification of the variables that influence football predictions.

4.5.1 Naïve Bayes

Naïve Bayes is a generative, probabilistic classifier that applies Bayes’ theorem while treating every predictor as conditionally independent once the class label is fixed. For an n -dimensional feature vector $\mathbf{x} = (x_1, \dots, x_n)$ and a class C_k , the posterior probability is

$$P(C_k | \mathbf{x}) = \frac{P(C_k) \prod_{i=1}^n P(x_i | C_k)}{\sum_j P(C_j) \prod_{i=1}^n P(x_i | C_j)}, \quad (3)$$

where $P(C_k)$ is the prior probability of class C_k , $P(x_i | C_k)$ is the likelihood of feature x_i given that class, and the denominator rescales the numerators so that the resulting posteriors sum to one (Wickramasinghe and Kalutarage, 2021).

It is assumed that conditional independence collapses the otherwise intractable joint likelihood $P(\mathbf{x} | C_k)$ into a product of one dimensional terms. Consequently, fitting the model requires only frequency counts in the categorical case. Because no iterative optimisation is needed, Naïve Bayes can be trained in seconds, even on big data with tens of thousands of dimensions.

4.5.2 Logistic Regression

Logistic regression uses a sigmoid link function which relates a linear score for each outcome to its probability, ensuring that predicted values are between 0 and 1 and that the probabilities for all classes sum up to one. In the general K -class (multinomial) case, separate linear predictors are assigned to each class and convert the set of scores to probabilities using a soft-max transformation (Hastie et al., 2023):

$$\Pr(Y = C_k | \mathbf{x}) = \frac{\exp(\beta_{0k} + \beta_k^\top \mathbf{x})}{\sum_{j=1}^K \exp(\beta_{0j} + \beta_j^\top \mathbf{x})}, \quad k = 1, \dots, K. \quad (4)$$

Here $\mathbf{x}$ is the feature vector, β_{0k} is an intercept, and β_k collects the coefficients for class C_k . Each coefficient $\beta_{j,k}$ describes how a one unit change in predictor X_j shifts the log odds in favour of class C_k relative to the others, while the denominator guarantees that all class probabilities sum to one.

4.5.3 Random Forest

Random Forest (RF) is an ensemble based classifier that grows M fully developed decision trees on bootstrap samples of the training data. At every node, each tree chooses its split from a random subset of predictors, adding two independent sources of randomness. For a feature vector x the ensemble prediction is

$$\hat{y}(x) = \frac{1}{M} \sum_{m=1}^M T_m(x), \quad (5)$$

where $T_m(x) \in \{1, \dots, k\}$ is the class label produced by the m^{th} tree in a k -class problem, M is the number of trees, and $\hat{y}(x)$ is the forest's final output, realised as a simple majority vote across the k categories (Breiman, 2001). Equivalently, one may form class probability estimates $\hat{p}_k(x) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}[T_m(x) = k]$ and predict the class with the highest $\hat{p}_k(x)$. These soft votes are useful in getting the predicted probabilities of each class, which helps to contradict bookmakers' allocated odds for a given match.

Because each tree omits roughly one third of the instances, the corresponding out-of-bag (OOB) examples furnish an internal, nearly unbiased estimate of the generalisation error. For multiclass tasks, the OOB framework extends naturally: a confusion matrix can be accumulated from OOB predictions and summarised by macro-averaged F_1 . Theoretically, the gap between the OOB error and the true forest error decreases exponentially fast as both the sample size n and the forest size M grow (Biau and Scornet, 2016).

4.5.4 Extreme Gradient Boosting (XGBoost)

XGBoost builds an additive ensemble of decision trees in a stage wise manner. At iteration k , the algorithm fits a new tree $f_k \in \mathcal{F}$ to the negative gradients (residuals) of the current model, thereby reducing the residual error. Two explicit regularisers- an ℓ_2 penalty λ on leaf weights and a per leaf cost γ -restrain model complexity and control over-fitting (Chen and Guestrin, 2016).

Mathematically, for a football match with feature vector $\mathbf{x}_i$ the final probability estimate is

$$\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i), \quad f_k \in \mathcal{F}. \quad (6)$$

While the model parameters are obtained by minimising the regularisation equation given as

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_{k=1}^K (\gamma T_k + \frac{\lambda}{2} \|\mathbf{w}_k\|_2^2), \quad (7)$$

where ℓ is the multinomial log loss for $\{\text{home-win, draw, away-win}\}$, T_k is the number of leaves of tree k and $\mathbf{w}_k$ its leaf scores. During training, a split is accepted only if it lowers equation (7), ensuring each added branch improves predictive accuracy while minimizing overfitting. XGBoost was chosen because of its strong predictive performance history and capacity to identify complex nonlinear relationships that may exist in football data (such as head-to-head records and team form combinations that jointly affect results).

4.5.5 Support Vector Machine

Support vector machine (SVM) learning seeks the hyperplane that maximises the margin between classes in a transformed feature space induced by a kernel function (Cortes and Vapnik, 1995). In a K -class setting ($K = 3$ for *home win*, *draw*, *away win*) a K one-vs-rest classifiers was used to predict

$$\hat{y}(x) = \arg \max_{k \in \{1, \dots, K\}} f_k(x), \quad (8)$$

where $f_k(x)$ is the signed distance given by the k -th binary SVM (Weston and Watkins, 1999). Posterior probabilities can be obtained by fitting a sigmoid to the decision values, yielding calibrated outputs suitable for log loss evaluation (Platt, 1999).

4.6 Model Evaluation

To assess the performance of the classification models, several standard evaluation and calibration metrics were employed. These metrics and their formal definitions are as follows:

- **Accuracy:** Accuracy measures the proportion of correct predictions among all predictions made. Let N_{correct} be the number of correct predictions and N_{total} be the total number of predictions (Géron, 2019). Then

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}}. \quad (9)$$

- **Precision:** Precision (also known as positive predictive value) represents the fraction of instances predicted as positive that are actually positive. Let TP and FP denote the number of true positives and false positives, respectively (Géron, 2019). Precision is then computed as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (10)$$

- **Recall:** Recall (also referred to as sensitivity or true positive rate) is the fraction of actual positive instances that the model correctly identifies. Let TP and FN denote the number of true positives and false negatives, respectively (Géron, 2019). Recall is given by

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (11)$$

- **F1-Score:** The F1-score is the harmonic mean of precision and recall, providing a single metric that balances both (Géron, 2019). It is calculated as

$$F1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (12)$$

- **ROC-AUC:** The Receiver Operating Characteristic (ROC) curve plots the true positive rate against the false positive rate at various classification thresholds. The area under this curve (ROC-AUC) summarises the model's ability to distinguish between the classes across all thresholds. An ROC-AUC of 1 indicates perfect class separation, whereas a value of 0.5 indicates performance equivalent to random guessing (Fawcett, 2006).
- **Balanced Accuracy:** Balanced accuracy is defined as the average of recall (sensitivity) over all classes. In a multiclass setting with K classes, if Recall_k denotes the recall for

class k , then

$$\text{Balanced Accuracy} = \frac{1}{K} \sum_{k=1}^K \text{Recall}_k. \quad (13)$$

For a binary case, this reduces to the average of sensitivity and specificity. Balanced accuracy is especially useful when dealing with imbalanced datasets ([Chicco and Jurman, 2020](#)).

- **Log Loss** (Cross-Entropy Loss): Log loss quantifies the performance of a classifier by penalising incorrect “soft” probability estimates. Let N be the total number of samples, and for each sample i , let $p_{i,c}$ be the predicted probability of the true class c , and let $y_{i,c}$ be an indicator (1 if sample i belongs to class c , else 0). Then

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^K y_{i,c} \log(p_{i,c}). \quad (14)$$

Lower log-loss indicates better calibrated probability estimates ([Chicco and Jurman, 2020](#)).

- **Brier Score**: The Brier score measures the mean squared error between the predicted probability distributions and the true one-hot encoded labels. Using the same notation as for log loss, the Brier score is

$$\text{Brier Score} = \frac{1}{N} \sum_{i=1}^N \sum_{c=1}^K (p_{i,c} - y_{i,c})^2. \quad (15)$$

A lower Brier score indicates that predicted probabilities are closer to the actual outcomes ([Guo et al., 2017](#)).

4.7 Methods to Uncover Potential Bookmakers' Manipulations

This section used the test data, bookmakers' odds, and the models' predictions to do a contrast study on the odds setting and models' performance.

4.7.1 Bookmakers' Vig (Margin)

The *vig* (or *overround*) is the margin built into the bookmaker's odds by which the sum of the implied probabilities of all possible outcomes exceeds 100%. It represents the bookmaker's expected profit margin on a "fair" market. For a three-way market (Home, Draw, Away) with decimal odds o_H , o_D , and o_A , the implied probability of each outcome is $p_i = \frac{1}{o_i}$, and the match vig is

$$\text{Vig}_{\text{match}} = \sum_{i \in \{H,D,A\}} p_i - 1 = \frac{1}{o_H} + \frac{1}{o_D} + \frac{1}{o_A} - 1. \quad (16)$$

This ensures that the bookmaker always secures a certain percentage profit, no matter the outcome of the match.

4.7.2 Identification of Potential Odd Manipulations

The logistic regression model, the random forest ensemble, and the extreme gradient boosting model will all give implied match odds. Those model-derived odds are then compared to the reported odds of each bookmaker individually. The analysis includes side-by-side graphs of each model's predicted odds and the bookmaker's odds for each matchup, emphasising the games with the biggest price differences. Each of these high discrepancy incidents is thoroughly examined to determine the underlying causes, which could be statistical limits, last minute information such as player injuries or lineup changes, or purposeful odds adjustments by the bookmaker to influence betting behaviour. This method not only identifies the areas where model predictions and market prices deviate most dramatically, but it also identifies the causes of such disparities.

4.7.3 Profit Simulation Using Model Predictions

The models' labelled predictions were compared with the actual match results to see the outcome if one uses any of the models to place bets. To do this, a 2-euro simulation bet per match was done, and the total profit or loss was calculated for each model and each bookmaker. To get the

profit or loss (π), the following was used:

$$\pi = \begin{cases} s(o - 1), & \text{if } \hat{y} = y \text{ and } o > 1, \\ -s, & \text{otherwise} \end{cases} \quad (17)$$

where

- $s = 2$ is the fixed stake,
- $o = o_{b,m}$ are the decimal odds offered by bookmaker (b) for the predicted outcome $m \in \{\text{H}, \text{D}, \text{A}\}$,
- $\hat{y} \in \{0, 1, 2\}$ is the model's predicted label (0=Draw, 1=Home, 2=Away),
- $y \in \{0, 1, 2\}$ is the actual result.

5 Results

5.1 Data Exploration

FTR	Percentage (%)
H (Home win)	43.60
A (Away win)	33.24
D (Draw)	23.17

Table 2: Percentage of Full Time Result categories

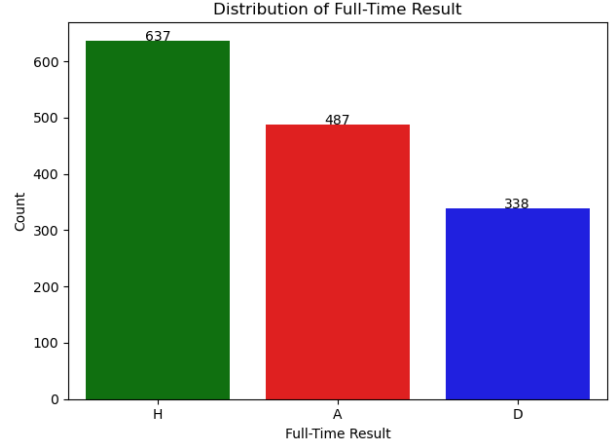


Figure 1: Distribution of Full Time Result

The response variable FTR (Figure 1) is a three level categorical outcome indicating whether the home team won (H), the away team won (A), or the match ended in a draw (D). In the dataset of 1,390 fixtures, home wins account for 43.60% of matches, away wins for 33.24%, and draws for 23.17%. This class imbalance, particularly the relative rarity of draws, motivated the use of targeted oversampling to ensure that models learn to recognise all outcomes with comparable effectiveness.

Figure 2 presents the five-season-long distribution of match outcomes for all twenty-two clubs that appeared in that period. Split into two groups of eleven, the top panel shows that elite teams like Club Brugge (96 wins, 40 draws, 25 losses) and Antwerp (88 wins, 36 draws, 37 losses) far outpace mid-table sides such as Charleroi (62 wins, 41 draws, 58 losses) or Genk (79 wins, 36 draws, 46 losses), while relegation candidates like Beerschot VA managed only 18 wins against 41 losses. In the bottom panel, Standard (60 wins, 44 draws, 57 losses) and St. Gilloise (68 wins, 18 draws, 12 losses) lead their group, whereas Mouscron recorded 19 draws, 16 wins, and 28 losses. Draws are most common among mid-table teams, who average around 35 to 45 draws, highlighting their role as crucial “points salvaged” in tight campaigns, while the very strongest and weakest sides register fewer draws, reflecting a tendency either to decisively win or lose.

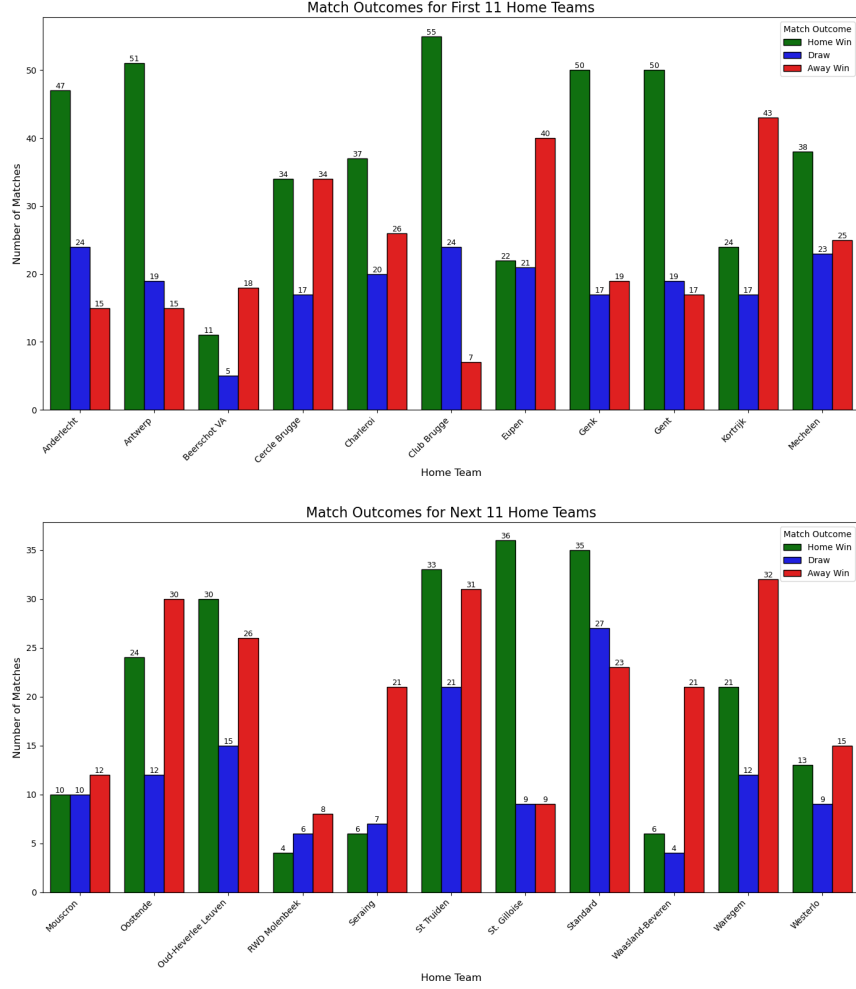


Figure 2: Distribution of Match Outcomes per Team (top and bottom panels).

5.2 Feature Engineering and Feature Selection

5.2.1 Feature Engineering (K-means Clustering)

The elbow method (Figure 3) was used to graph inertia versus candidate k values to determine the appropriate number of groups. Figure 4 shows the match level feature space onto its first two principal component axes and color codes every fixture by the centroid assigned by $k = 3$. The sky blue cluster (upper right) is tight and distant from the grand mean, signalling fixtures where multiple attacking metrics favour the home side (confident host). The blue cluster (lower right) lies closer to the axis that loads heavily on away-team shot volumes, reflecting matches in which a surging visitor tilts the balance. The green cluster (left) straddles both axes with greater

spread, consistent with lower-intensity, evenly poised fixtures, hence the balanced contest label.

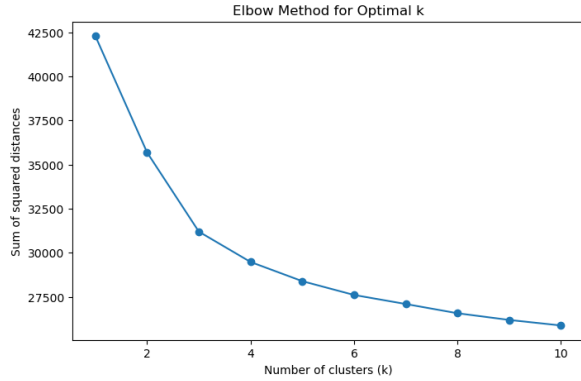


Figure 3: Elbow method for Optimal K in K-means Clustering

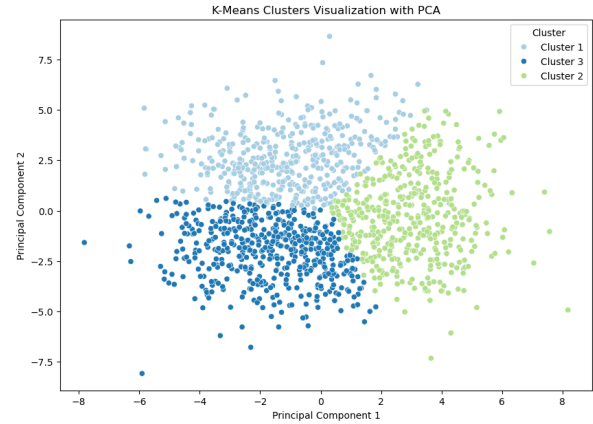


Figure 4: Visualization of the Match Clusters Assignments

Table 7 in the appendix confirms this visual reading by listing how often each club's games fall into each cluster. Persistent title contenders Anderlecht (75 appearances), Club Brugge (73), Genk (71), and Gent (71) dominate Cluster 1, reinforcing its confident-host character. Cluster 2 shows a strong visitor influence led by Antwerp (88), Gent (85), Club Brugge (85), Genk (76), and Anderlecht (72), reflecting matches in which a surging away side tilts the balance. Cluster 3 is populated chiefly by more modest clubs: Eupen (94), St. Truiden (88), and Kortrijk (82), illustrating balanced contests where neither side enjoys a clear statistical edge.

5.2.2 Feature Selection

The Spearman correlation heatmap (Figure 5) indicates that there was severe multicollinearity because the constituent sub ratings for both home and away teams (attack, midfield, and defence) showed pairwise correlations above 0.80. In the final model formulation, all individual sub-rating features were removed and only the two aggregate variables, `Overall Rating_Home` and `Overall Rating_Away` were kept to avoid duplicate predictors.

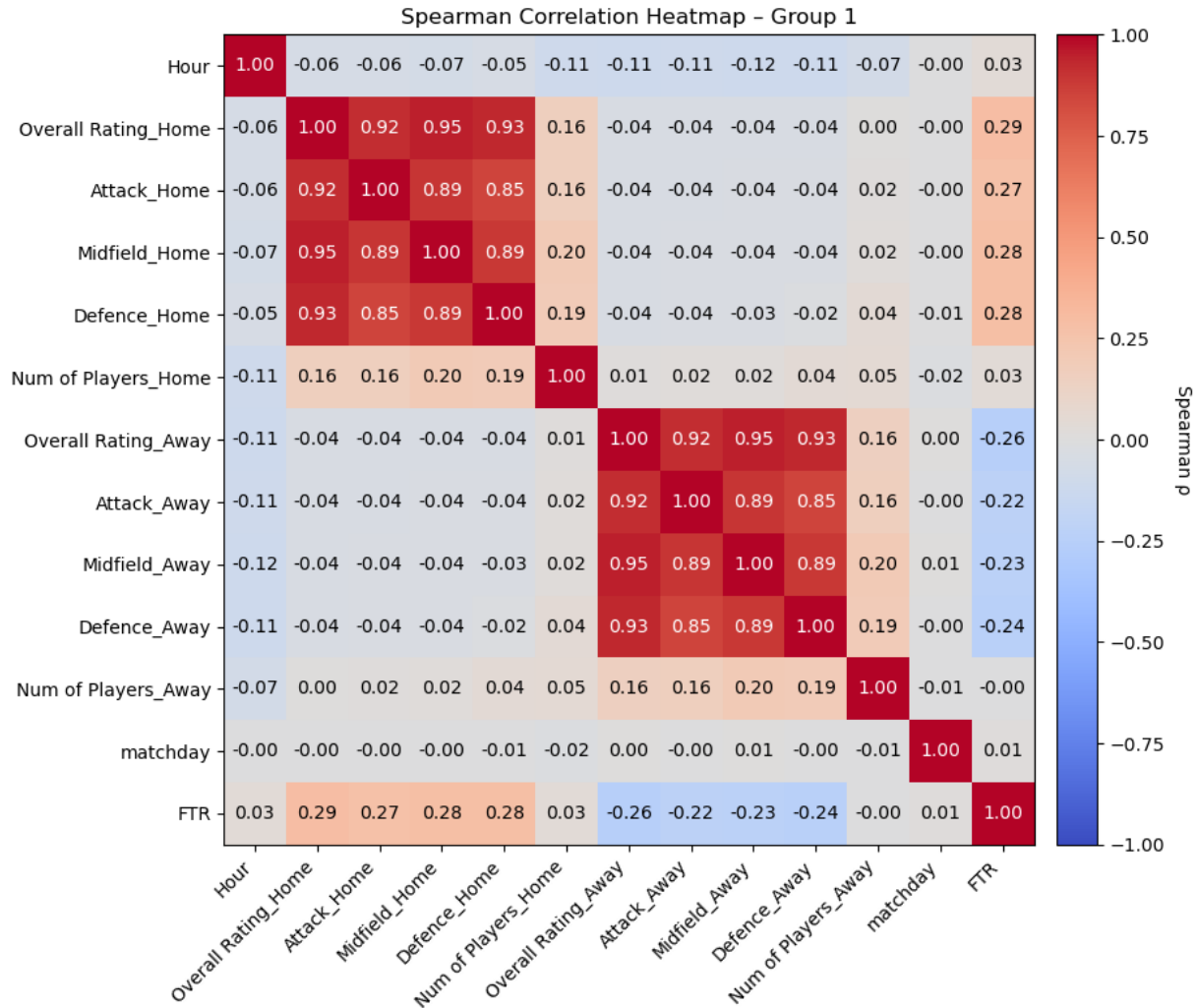


Figure 5: Spearman Correlation Heatmaps for Feature Selection

5.3 Model Results

The five examined models perform similarly (Table 3), with macro-averaged ROC AUC scores ranging from 0.615 to 0.696. Random Forest achieves the highest discriminative performance ($\text{AUC} \approx 0.696$), followed closely by Logistic Regression ($\text{AUC} \approx 0.691$). Both indicate that the engineered features capture a fairly strong, linearly separable signal, despite Naive Bayes’s simplistic independence assumptions. The support vector machine lags noticeably ($\text{AUC} \approx 0.615$), suggesting that its RBF (Radial Basis Function) decision boundary struggles to capture the multiclass structure under this feature set.

In terms of overall accuracy and weighted F1 score, the five models occupy a narrow band. Support Vector Machine records the lowest accuracy (approximately 45.2%) and F1 score (approximately 44.8%), while Random Forest attains the highest accuracy ($\approx 54.0\%$) and the highest F1 score for the majority classes ($\approx 49.3\%$). Logistic Regression also performs well, with accuracy $\approx 51.5\%$ and F1 score $\approx 49.7\%$. Naive Bayes yields accuracy $\approx 51.0\%$ and F1-score $\approx 50.0\%$, reflecting that class weighting and oversampling produced roughly symmetric performance across classes

Calibration metrics further differentiate the models. Random Forest has the lowest log loss (≈ 0.986) and the lowest Brier score (≈ 0.590), indicating better calibrated probability estimates. Logistic Regression follows with log loss ≈ 1.039 and Brier score ≈ 0.627 , while Naive Bayes has the highest log loss (≈ 1.512) and Brier score (≈ 0.722), reflecting overly confident but incorrect probability assignments.

Table 3: Comparison of Model Performance Metrics on the Test Set (Transposed)

Metric	Logistic Regression	Random Forest	SVM	XGBoost	Naive Bayes
Accuracy	0.515	0.540	0.452	0.485	0.510
Balanced Accuracy	0.502	0.501	0.437	0.465	0.502
Precision	0.529	0.541	0.450	0.479	0.524
Recall	0.515	0.540	0.452	0.485	0.510
F1 Score	0.497	0.493	0.448	0.476	0.500
ROC AUC	0.691	0.696	0.615	0.666	0.662
Log Loss	1.039	0.986	1.088	1.063	1.512
Brier Score	0.627	0.590	0.652	0.638	0.722

The per-class breakdown (Table 4) and the confusion matrices (Figure 10) reveal that all models struggle most with predicting draws, even after targeted oversampling. Logistic Regression attains a draw precision 0.43 and recall of 0.20, thereby missing 80% of actual draws. Naive Bayes falls between these extremes, with a draw precision 0.41 and a recall 0.25. By contrast, all models perform substantially better on home win and away win outcomes.

Aggregating across classes, Random Forest emerges as the top performer, achieving the highest overall accuracy ($\approx 53.97\%$), balanced accuracy ($\approx 50.07\%$), lowest log loss (≈ 0.986), lowest Brier score (≈ 0.590), and best macro AUC (≈ 0.696). Logistic Regression and Naive Bayes deliver similar performance both achieve accuracy around 51.0%, balanced accuracy around 50.0%, and AUC around 0.691 (Logistic Regression) to 0.663 (Naive Bayes), demonstrating that even simple linear or generative models capture much of the signal when features are well engineered.

Table 4: Per-Class Performance of All Models on the Test Set

Model	Class	Precision	Recall	F1-Score	Support
Logistic Regression	Draw (0)	0.43	0.20	0.27	65
	Home Win (1)	0.65	0.56	0.60	108
	Away Win (2)	0.43	0.74	0.54	66
Random Forest	Draw (0)	0.50	0.09	0.16	65
	Home Win (1)	0.63	0.71	0.67	108
	Away Win (2)	0.44	0.70	0.54	66
Support Vector Machine	Draw (0)	0.28	0.23	0.25	65
	Home Win (1)	0.57	0.52	0.54	108
	Away Win (2)	0.43	0.56	0.49	66
XGBoost	Draw (0)	0.28	0.20	0.23	65
	Home Win (1)	0.62	0.57	0.60	108
	Away Win (2)	0.44	0.62	0.52	66
Naive Bayes	Draw (0)	0.41	0.25	0.31	65
	Home Win (1)	0.65	0.55	0.59	108
	Away Win (2)	0.43	0.71	0.54	66

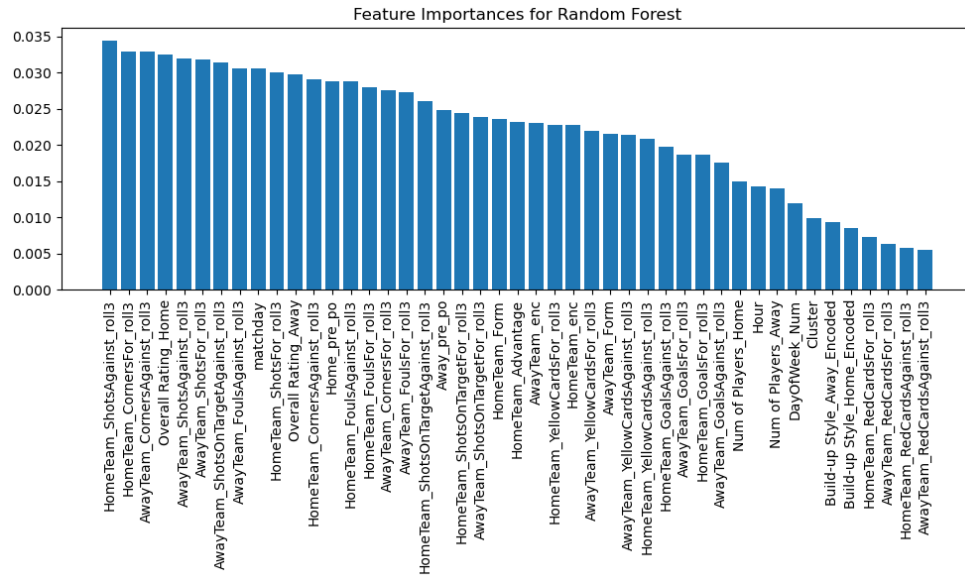
Overall, targeted oversampling of draws enabled all classifiers to learn draw specific patterns, but with noticeable precision recall trade offs. Ensemble methods such as Random Forest and

linear models such as Logistic Regression emerge as the most robust options, balancing false positives and negatives across match outcome classes while yielding the best calibration (lowest log loss and Brier score).

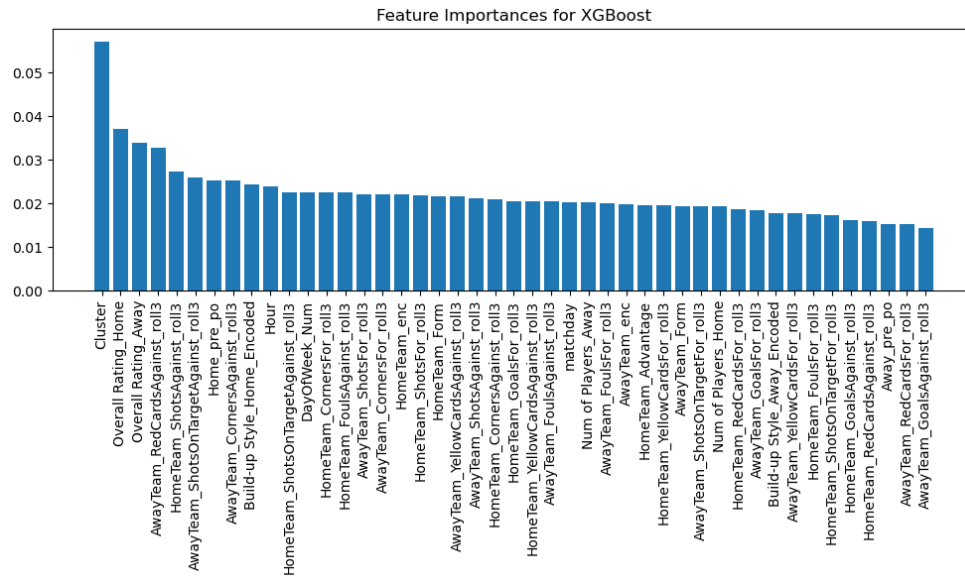
5.3.1 Feature Importance Results

Figures 6a and 6b show the top predictors for Random Forest and XGBoost, respectively. In the Random Forest model, the two most important features are **HomeTeam_ShotsAgainst_roll3** (shots conceded by the home side over their last three matches) and **HomeTeam_CornersFor_roll3** (corners earned by the home side over their last three matches), indicating that recent defensive solidity and chance creation at home are primary drivers of win probability. The next two features are **Overall_Rating_Home** and **Overall_Rating_Away** (each team’s current aggregated squad rating), confirming that higher rated squads tend to dominate. The fifth highest feature is **AwayTeam_ShotsFor_roll3** (shots taken by the away side over their last three fixtures). Together, these top five variables underscore that short term defensive performance (minimising shots conceded), offensive chance creation (corners and shots for), and both teams’ overall quality carry the most weight in the Random Forest’s estimates. Other highly ranked inputs (such as rolling fouls against, **ShotsOnTargetAgainst** metrics, and form related ratings) further highlight the importance of recent defensive form, chance creation ability, and momentum.

By contrast, XGBoost places **Cluster** (a pre computed grouping variable that captures tactical or league level similarity) as its single most important predictor, suggesting that grouping teams by playing style or competitive context provides a powerful high level signal. The next two features are once again **Overall_Rating_Home** and **Overall_Rating_Away**, followed by **AwayTeam_ShotsAgainst_roll3** (shots conceded by the away side over three games) and **HomeTeam_ShotsAgainst_roll3**. In other words, XGBoost also values aggregate squad quality (static ratings for home/away) and short-term defensive solidity (rolling “shots against”), but it sits directly behind a tactical clustering variable. Other top dozen inputs in XGBoost include “Build-up Style” (possession/formation encoding), **HomeTeam_CornersFor_roll3**, and **HomeTeam_FoulsAgainst_roll3**, again emphasising how tactical grouping, recent form, and defensive pressure influence match outcomes.



(a) Random Forest Feature Importances



(b) XGBoost Feature Importances

Figure 6: Feature importance plots for (a) Random Forest and (b) XGBoost.

5.4 Calculation of Bookmakers' Average Profit Margin (Vig)

Table 5 presents the average vigorish (house edge) charged by each bookmaker on full-time result markets, computed over the test dataset. A higher vig represents a larger implicit commission, reducing bettors' expected returns. Bet Victor (VC) imposes the highest mean vig of 7.03%, followed by Bet365 (B365) at 6.63% and Betbrain (Avg) at 6.20%. Bet and Win (BW) comes next with 5.78%, while Pinnacle (PS) offers more favourable odds with a vig of 4.18%. MaxBet (Max) provides the thinnest margin at only 1.94%. Overall, this analysis shows that, regardless of the match outcome, bookmakers secure an average profit from each stake by carefully adjusting their odds.

Table 5: Average Vigorish (House Edge) by Bookmaker on the Test Data

Bookmaker	Average Vig
Bet365 (B365)	0.0663
Bet and Win (BW)	0.0578
Pinnacle (PS)	0.0418
Bet Victor (VC)	0.0703
MaxBet (Max)	0.0194
Betbrain (Avg)	0.0620

5.5 Plots of Bookmaker Odds versus Model Odds

Figure 7 shows comparative line graphs of the 'fair' odds derived from Random Forest against Bet Victor's (VC) published market odds for home wins (top), draws (middle), and away wins (bottom) in the test set. In most cases, the two series track closely, indicating that the model successfully captures the prevailing market signal. However, notable divergences appear at specific match indices. For instance, in the Home Win panel, around match index 175, the RF fair odds rise toward approximately 5, whereas VC maintains odds near 9, suggesting that the model assigns a substantially lower probability to an upset than the bookmaker. Likewise, in the Away Win panel, a spike near 20 around index 180 implies that VC is overpricing a heavy underdog relative to the model's projection.

Table 11 in the appendix documents the five most fixtures in which VC's odds exceed RF's

implied odds by the largest margin (VC-RF) in each market. Each row isolates a single match and outcome, and the column Δ (VC-RF) quantifies that difference. Positive values in Δ highlight potential “value” bets where VC offers more generous odds than the model would imply. For example, on 2023-12-23 (Eupen vs. St. Gilloise, Home Win), RF implies odds of 3.33 ($\approx 30\%$ implied probability), while VC posts 12.00 ($\approx 8\%$ implied probability), yielding $\Delta = +8.67$. Conversely, small or negative Δ values would identify matches where VC’s quoted odds are tighter than the RF expectation, potentially indicating overpricing by the bookmaker. By combining the visual insights from Figure 7 with the precise tabular extremes in Table 11, one can systematically pinpoint where model based probabilities and bookmaker margins diverge most sharply, thereby uncovering the strongest value opportunities (large positive Δ) as well as potential mispricings where the market may be too conservative (negative or small Δ).

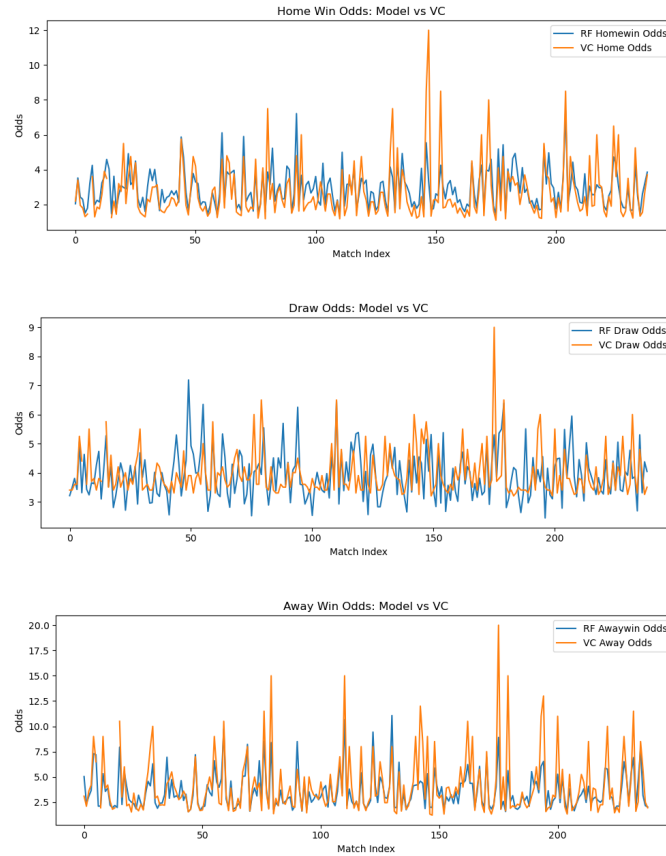


Figure 7: Plot of selected bookmaker odds versus random forest generated odds for home wins (top), draws (middle), and away wins (bottom).

5.6 Profit Simulation Results by Model and Bookmaker

The profit simulation (Table 6 and Figure 8), based on a consistent €2 stake per match, reveals notable differences in how each model performs across various bookmakers. Naïve Bayes achieved the highest returns: €22.32 on BetBrain (Avg), driven largely by strong performance with MaxBet (€40.74) and Pinnacle (€28.24), as its tendency to predict draws (which carry higher odds) paid off. Logistic Regression also generated solid profits, Betbrain €11.12, with a peak of €28.60 at MaxBet. Random Forest returned modest gains overall (€6.66), but lost money at Bet Victor (−€0.84). In contrast, XGBoost and Support Vector Machine both lost money on Betbrain (−€6.68 and −€17.40, respectively), with XGBoost only profitable at MaxBet (€9.84) and Support Vector Machine losing across all six bookmakers (for instance, −€18.78 at Bet365 and −€20.76 at Bet Victor). These results indicate that predictive accuracy alone does not guarantee betting profitability; models that correctly identify higher odds outcomes (like draws) or align closely with specific bookmakers' pricing may yield better returns even if their overall accuracy is lower.

Table 6: Total Profit (€) on the Test Set by Model and Bookmaker (€2 per Bet)

Model	Avg	B365	BW	Max	PS	VC
Logistic Regression	11.12	10.88	15.36	28.60	17.12	0.92
Naive Bayes	22.32	22.30	26.28	40.74	28.24	6.92
Random Forest	6.66	4.76	10.90	22.36	12.00	−0.84
Support Vector Machine	−17.40	−18.78	−14.02	−0.20	−10.76	−20.76
XGBoost	−6.68	−6.66	−2.66	9.84	−0.48	−16.00

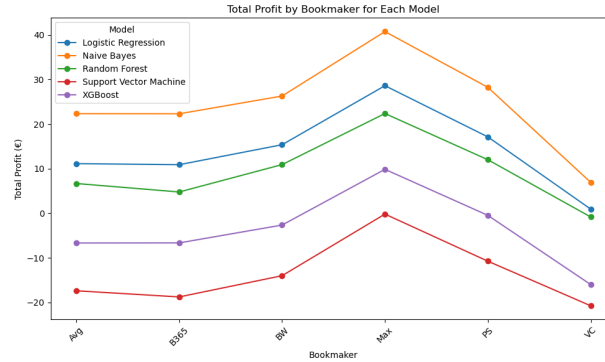


Figure 8: Plot of Total Profit in Euros on the Test Set by Model and Bookmaker

6 Discussion and Drawbacks of the Research

6.1 Discussion

The results of this study highlight both the potential and the limitations of using ML to predict football outcomes in a balanced and competitively unpredictable league such as the Belgian Jupiler Pro League. Across five seasons of match data, five distinct classification algorithms: Naïve Bayes, Logistic Regression, Random Forest, XGBoost, and Support Vector Machine were evaluated. While Random Forest emerged as the top performing model in terms of discrimination (macro-ROC AUC ≈ 0.696), accuracy ($\approx 54.0\%$), and probability calibration (log loss = 0.986, Brier score = 0.590), all models performed within a relatively narrow band of effectiveness. This convergence aligns with earlier research indicating that even advanced ML techniques often stop at accuracy levels between 50% and 55% for three-way football classifications ([Hubáček et al., 2019](#); [Holmes and McHale, 2024](#)).

One of the most notable findings from the per-class evaluation was the consistent difficulty models faced in accurately predicting draws. Even after oversampling techniques were applied to rebalance class frequencies, the recall for the draw category remained below 25% across all models. This limitation highlights the intrinsic volatility of football matches, especially in cases where teams are evenly matched and the game lacks a dominant narrative. Prior studies have suggested that draws are both statistically rarer and more context dependent, which likely contributes to the low recall despite balanced training data ([Constantinou et al., 2013](#)).

Feature importance analysis provided actionable insights into the predictive variables most associated with match outcomes. Both Random Forest and XGBoost models ranked recent defensive stability, particularly the three match rolling average of shots conceded, as critical. This suggests that short-term form indicators, more so than static team attributes, may carry substantial weight in shaping match probabilities. Additionally, the inclusion of squad rating variables confirms prior literature ([Kovalchik, 2020](#)) emphasizing that overall team quality, as captured through ratings from platforms like Sofifa, plays a central role in outcome prediction. XGBoost also assigned high importance to a custom “Cluster” variable derived from K-means grouping of match level features. This clustering likely captured latent tactical styles or league context patterns, supporting the growing notion that higher level contextual embeddings can enhance football forecast accuracy ([Luiz et al., 2024](#)).

Despite moderate predictive success, model based probability estimates revealed several dis-

crepancies when compared with actual bookmaker odds. Bookmakers, particularly Bet Victor and Bet365, imposed average vigorish margins exceeding 6.5%, reinforcing the idea that betting markets are not entirely efficient nor purely reflective of on field dynamics. Models such as Naïve Bayes and Logistic Regression identified several "value bets" situations where model implied odds significantly diverged from bookmaker odds. In some of these cases, placing model based bets would have led to considerable simulated profits, particularly under MaxBet's lower vig regime. For example, Naïve Bayes returned €40.74 in simulated profit when betting at MaxBet, illustrating that simplicity in model structure does not preclude market outperformance when probabilistic alignment is strong. However, profitability did not consistently align with accuracy. XGBoost and Support Vector Machine, despite moderately good calibration, returned losses in most scenarios, suggesting that accuracy alone is insufficient for economic gains in betting markets. This paradox mirrors findings from [Myrsten, 2021](#), who argued that model profitability depends not just on precision but also on identifying when market odds misprice outcomes, particularly undervalued longshots such as draws.

Importantly, this research holds broader implications for market integrity and public awareness. By systematically comparing model generated probabilities with bookmaker odds, the study exposes embedded biases in odds setting. Bookmakers appear to manipulate odds not only to reflect expected outcomes but also to exploit bettor behavior ([Levitt, 2004](#)), especially around popular teams and home advantage. This "odds shading" can mislead bettors, especially when compounded by a lack of transparency around vig and pricing rationale. Accordingly, this study does not merely aim to enhance prediction but also to sensitize bettors to the structural disadvantage they face; a theme echoed in recent calls for more ethical, responsible betting ecosystem ([Vieira et al., 2023](#)).

In conclusion, while machine learning offers valuable tools for analyzing match outcomes and bookmaker behavior, the inherent randomness of football and the profit driven architecture of betting markets present persistent barriers. Still, with well engineered features and careful calibration, standard models can not only improve predictive insight but also serve as watchdogs for market fairness and educational instruments for bettor awareness.

6.2 Drawbacks of the Research

The study was methodologically sound, but various constraints limit its generalisability. The data set is limited to five seasons (2019/20 to 2023/24) from a single tournament, which saw structural disturbances like COVID-19 postponements and format changes. The sample size of approximately 1400 fixtures limits statistical power, and the use of match level aggregates obscures player-specific variation and tactical complexity, limiting model explanatory depth.

Second, modelling simplifications may generate systematic bias. Duplicating previous draws and away wins with random oversampling alters the target distribution’s natural base rates and may inflate apparent calibration when the model is used on imbalanced data. Profits reported here are illustrative and not directly deployable, as they are based on a stylised 2 euros flat stake approach in the 1X2 market, excluding bankroll management restrictions, transaction expenses, and bookmaker-imposed stake limits.

The feature set does not include real-time and micro level factors that bookies use immediately, such as confirmed starting XI, last-minute injuries, weather conditions, and in-play occurrences. The lack of high frequency data helps explain the miscalibration between model-implied and market implied probability, highlighting the inherent disadvantage of pre-match models. Addressing these omissions and methodological constraints may offer more robust and actionable insights.

7 Ethics, Societal Relevance, and Stakeholder Awareness

7.1 Ethics

One of the primary ethical challenges lies in maintaining transparency when identifying discrepancies between predicted probabilities and bookmaker odds. This research is designed to promote awareness and responsible gambling, making sure that it does not encourage risky behaviours or create misunderstandings about the nature of betting markets. While analysing bookmaker margins (the "vig"), the research avoids drawing conclusions that could unfairly damage the reputation of bookmakers. The analysis is intended to provide a clear understanding of betting mechanics, empowering bettors to make informed decisions without encouraging excessive or harmful betting practices.

Another critical ethical dimension concerns the calculation of feature importance in predictive models. The insights gained through this analysis are shared responsibly to prevent misuse of data while enabling stakeholders to make better decisions. Ensuring that feature importance is presented ethically prevents the potential exploitation of sensitive data while also empowering individuals and organisations to make well informed choices that contribute to fairer outcomes in football prediction and betting.

7.2 Societal Relevance

Sports betting has grown into a global phenomenon, with millions of individuals participating each year. However, many bettors are unaware of how bookmakers incorporate a profit margin into the odds, ultimately skewing their chances of winning. This research serves to educate bettors about these hidden costs, thereby encouraging more responsible betting behavior. By identifying and analyzing discrepancies between predicted probabilities and bookmaker odds, this study highlights areas where bettors are likely to misinterpret or overestimate their chances of success. These findings aim to support public awareness campaigns designed to combat gambling addiction and promote responsible gambling practices ([Hing et al., 2015](#)).

The research also addresses the societal challenge of gambling addiction. The analysis of probabilities and the identification of biases in bookmaker odds emphasize where misperceptions can lead to overconfidence among bettors. By spotlighting these issues, the study aims to contribute to initiatives focused on reducing problem gambling and increasing awareness of the risks associated with sports betting ([Hing et al., 2015](#); [Serby, 2015](#)). Furthermore, by analysing book-

maker margins, the research provides transparency in an industry often criticized for its lack of openness. This transparency allows individuals to engage in betting with a clearer understanding of the costs involved, fostering a more informed and healthier gambling culture.

7.3 Stakeholder Awareness

This research holds significant value for a range of stakeholders, each benefiting in unique ways. Bettors, as the primary stakeholders, are affected by the hidden margins embedded in bookmaker odds that favor bookmakers. By comparing predicted probabilities with bookmaker odds, this study helps bettors identify potentially unfair practices, allowing them to make better informed decisions. For researchers and data scientists, the study advances the field of sports analytics and machine learning. By identifying key features that influence match outcomes, it offers new insights into the factors that drive prediction accuracy in football, contributing to the development of predictive models that are not only effective but also ethical and responsible.

Sports teams and coaches can also benefit from the research by gaining a better understanding of the factors influencing match results. Insights into features such as team dynamics and home advantage provide actionable data that can inform strategies, opponent scouting, and resource allocation. For example, the research identifies that defensive stability or recent momentum are significant predictors of success; teams may adjust their training and tactics to enhance their chances of winning. Bookmakers, although critiqued in this research, stand to gain from increased transparency. By clarifying how margins are calculated and identifying areas for improvement, bookmakers can enhance their credibility and create trust with bettors, helping to improve the relationship between bookmakers and their customers and creating a fairer and more competitive environment.

8 Conclusion and Future Research

8.1 Conclusion

Standard machine learning models trained on publicly accessible pre-match data can achieve bookmaker level discrimination ($AUC \approx 0.70$) while concurrently revealing significant price biases. Rolling defensive measures, corner productivity, and comprehensive squad ratings account for the majority of volatility; nonetheless, match outcomes remain difficult to predict reliably. Market analysis indicates that betting operators consistently incorporate a margin of 2 to 7% and occasionally adjust long shot prices beyond their fair value; strategic exploitation of these discrepancies results in modest, bookmaker specific profits, illustrating that successful wagering relies on targeted application of models rather than indiscriminate use. The probability difference framework provides regulators with a clear auditing tool, bettors with a structured value assessment, and bookmakers with a standard for refining odds. Future benefits will likely arise from enhanced context, verified line-ups, micro events, player tracking feeds, and from temporal frameworks capable of modelling team trajectories without artificial balancing. Incorporating these advancements inside responsible gambling safeguards is crucial. The statistical house edge, however minimal, continues to favour bookmakers and necessitates vigilance from all market players.

8.2 Future Research

The initial objective of future research should be to enhance the informational baseline that drives the models. The algorithms would be able to capture tactical complexity that is invisible in pre-match aggregates by incorporating confirmed line-ups, player level fitness and workload indicators, and spatial monitoring or event stream data. Combining these more detailed features with temporally explicit architectures, such as Bayesian dynamic ratings or recurrent neural networks, would enable team strengths to develop on a weekly basis and eliminate the necessity for synthetic oversampling. The portability of Belgium trained features and weightings would be further tested by replicating the process across leagues of contrasting manner (for example, Eredivisie and Ligue 1).

The model outputs in the regulatory and live wagering domains should be the focus of a second strand. The thin mispricing pockets documented here would be clarified as to whether they persist under increased information flow by extending probability calibration to in-play markets, where odds react in real time to goals, cards, and momentum shifts. Profitability estimates would be based on realistic trading constraints through parallel work on bankroll management rules and stake size limits. Finally, the alignment of technical advancements with consumer protection objectives could be achieved by translating model detected gaps into user facing dashboards and integrating odds movement anomaly detection with integrity analytics. This approach could promote responsible gambling and early warning for match fixing.

8.3 Softwares Used

The following software and tools were used to complete this Research:

- Python 3.11.7 for data analysis (exploration and model building)
- Github for Storage of the work and version control
- Microsoft Excel 2016 (Power Query) for data collection and analysis
- Overleaf was used to write the entire report

References

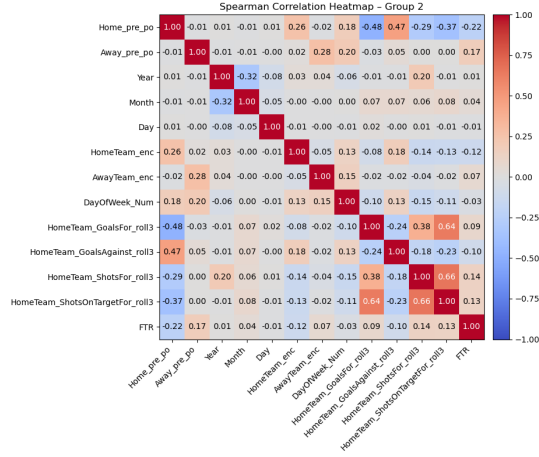
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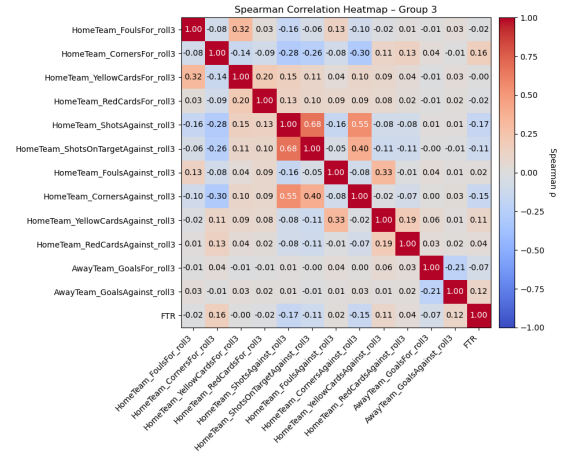
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Appendix

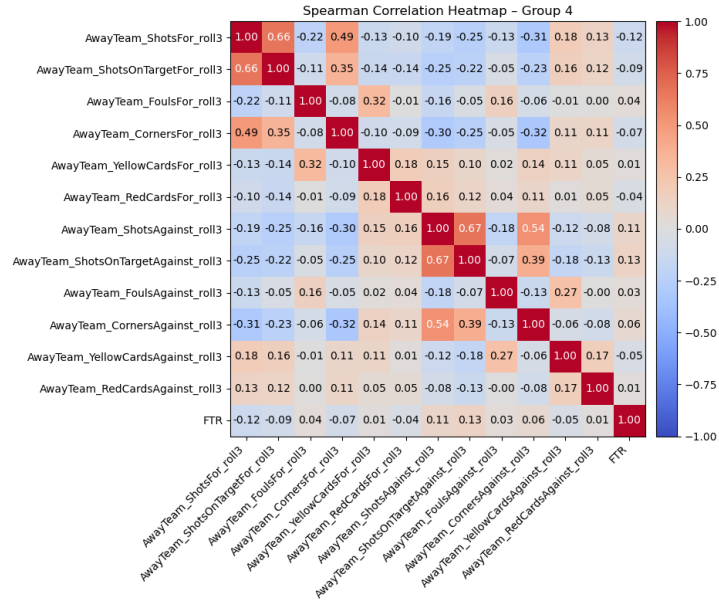
Plots



(a) Feature set 3

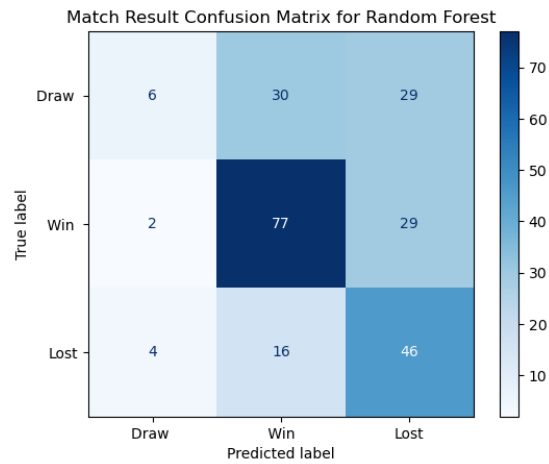


(b) Feature set 4

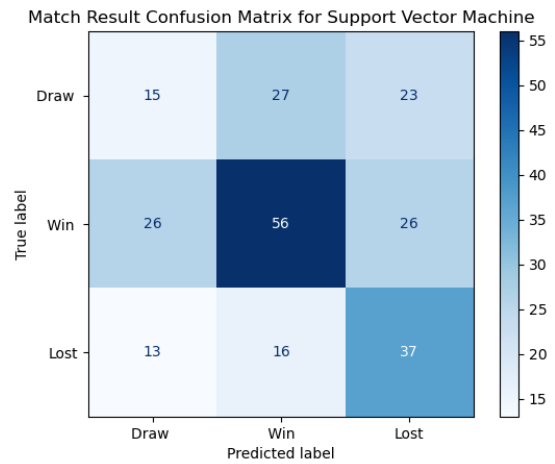


(c) Feature set 5

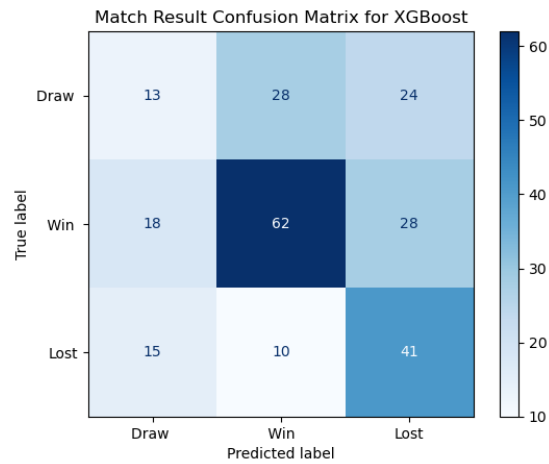
Figure 9: Correlation Heatmaps for Feature Selection: (a) Set 2, (b) Set 3, (c) Set 4



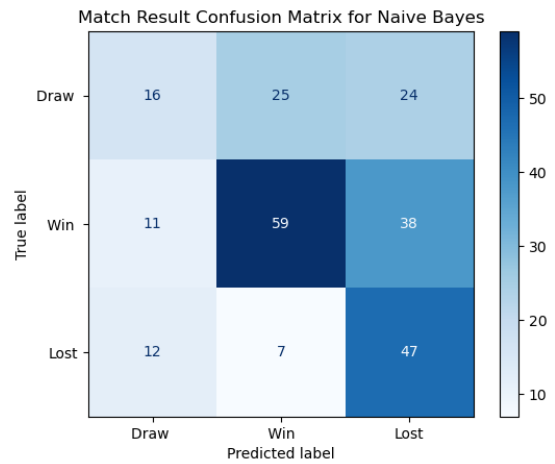
(a) Random Forest



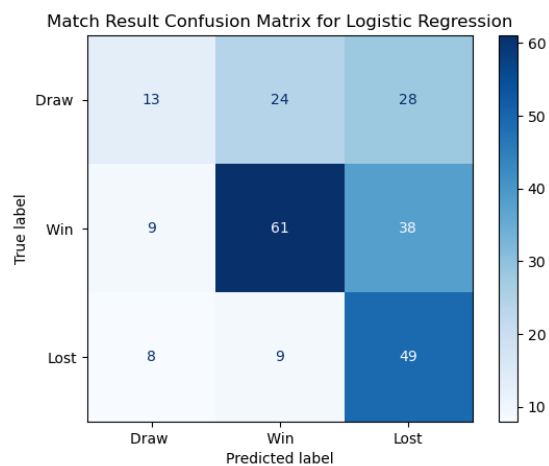
(b) Support Vector Machine



(c) XGBoost



(d) Naive Bayes



(e) Logistic Regression

Figure 10: Confusion matrices of all five models

Tables

Table 7: Team appearances in the three K-means clusters (updated)

Team	Cluster 1	Cluster 2	Cluster 3
Anderlecht	75	72	11
Antwerp	65	88	6
Beerschot VA	17	15	34
Cercle Brugge	48	41	69
Charleroi	49	52	58
Club Brugge	73	85	0
Eupen	32	32	94
Genk	71	76	11
Gent	71	85	2
Kortrijk	38	38	82
Mechelen	48	39	71
Mouscron	13	9	39
Oostende	31	26	71
Oud-Heverlee Leuven	35	30	66
RWD Molenbeek	7	5	17
Seraing	11	11	44
St. Truiden	41	30	88
St. Gilloise	39	57	0
Standard	49	45	65
Waregem	28	25	74
Waasland-Beveren	6	7	48
Westerlo	19	12	32

Click here to go back to K-means [Elbow plot](#).

Table 8: Correlation of Features with Target Variable FTR_encoded

		Feature	Corr.
Feature	Corr.		
		Build-up Style_Away_Encoded	-0.0013
Overall Rating_Home	0.2908	HomeTeam.YellowCardsFor_roll3	-0.0025
Away_pre_po	0.1729	Build-up Style_Home_Encoded	-0.0125
HomeTeam_Form	0.1695	HomeTeam.FoulsFor_roll3	-0.0164
HomeTeam_CornersFor_roll3	0.1573	AwayTeam.CornersAgainst_roll3	-0.0164
HomeTeam_Advantage	0.1505	AwayTeam.YellowCardsAgainst_roll3	-0.0555
HomeTeam_ShotsFor_roll3	0.1350	Cluster	-0.0493
AwayTeam_ShotsOnTargetAgainst_roll3	0.1331	AwayTeam_enc	-0.0491
HomeTeam_ShotsOnTargetFor_roll3	0.1311	AwayTeam.ShotsAgainst_roll3	-0.0518
AwayTeam_GoalsAgainst_roll3	0.1176	AwayTeam.ShotsOnTargetAgainst_roll3	-0.0572
AwayTeam_ShotsAgainst_roll3	0.1097	HomeTeam.ShotsAgainst_roll3	-0.1680
HomeTeam.YellowCardsAgainst_roll3	0.1095	HomeTeam.CornersAgainst_roll3	-0.1463
HomeTeam_GoalsFor_roll3	0.0943	AwayTeam_Form	-0.1380
AwayTeam_enc	0.0668	HomeTeam_Advantage	-0.1505
AwayTeam_CornersAgainst_roll3	0.0643	Overall Rating_Away	-0.2552
Month	0.0385	Home_pre_po	-0.2243
HomeTeam_RedCardsAgainst_roll3	0.0356	HomeTeam.RedCardsFor_roll3	-0.0196
AwayTeam_FoulsFor_roll3	0.0352	AwayTeam_RedCardsFor_roll3	-0.0385
AwayTeam_FoulsAgainst_roll3	0.0313	AwayTeam_GoalsFor_roll3	-0.0678
HomeTeam_FoulsAgainst_roll3	0.0204	AwayTeam_CornersFor_roll3	-0.0741
AwayTeam.YellowCardsFor_roll3	0.0132	AwayTeam_ShotsOnTargetFor_roll3	-0.0914
matchday	0.0092	HomeTeam_GoalsAgainst_roll3	-0.0965
AwayTeam_RedCardsAgainst_roll3	0.0082	HomeTeam_ShotsOnTargetAgainst_roll3	-0.1146
		HomeTeam_enc	-0.1221
		AwayTeam_ShotsFor_roll3	-0.1228

Click here to go back to feature selection [correlation plot](#).

Table 9: Feature Importances for Logistic Regression, Random Forest, and XGBoost

Feature	Logistic Regression	Random Forest	XGBoost
AwayTeam_CornersAgainst_roll3	0.029	0.032	0.029
AwayTeam_CornersFor_roll3	0.018	0.028	0.022
AwayTeam_Form	0.037	0.024	0.022
AwayTeam_FoulsAgainst_roll3	0.059	0.031	0.024
AwayTeam_FoulsFor_roll3	0.000	0.028	0.023
AwayTeam_GoalsAgainst_roll3	0.002	0.019	0.018
AwayTeam_GoalsFor_roll3	0.038	0.020	0.021
AwayTeam_RedCardsAgainst_roll3	0.037	0.007	0.023
AwayTeam_RedCardsFor_roll3	0.033	0.007	0.020
AwayTeam_ShotsAgainst_roll3	0.018	0.032	0.025
AwayTeam_ShotsFor_roll3	0.038	0.032	0.020
AwayTeam_ShotsOnTargetAgainst_roll3	0.000	0.032	0.026
AwayTeam_ShotsOnTargetFor_roll3	0.003	0.026	0.019
AwayTeam_YellowCardsAgainst_roll3	0.046	0.024	0.025
AwayTeam_YellowCardsFor_roll3	0.007	0.022	0.020
AwayTeam_enc	0.006	0.026	0.019
Away_pre-po	0.005	0.028	0.023
Build-up Style_Away_Encoded	0.024	0.011	0.021
Build-up Style_Home_Encoded	0.010	0.010	0.025
Cluster	0.035	0.007	0.025
HomeTeam_Advantage	0.017	0.024	0.022
HomeTeam_CornersAgainst_roll3	0.005	0.030	0.022
HomeTeam_CornersFor_roll3	0.088	0.032	0.023
HomeTeam_Form	0.027	0.023	0.026
HomeTeam_FoulsAgainst_roll3	0.014	0.029	0.025
HomeTeam_FoulsFor_roll3	0.006	0.030	0.023
HomeTeam_GoalsAgainst_roll3	0.007	0.020	0.019
HomeTeam_GoalsFor_roll3	0.002	0.019	0.018
HomeTeam_RedCardsAgainst_roll3	0.000	0.007	0.023
HomeTeam_RedCardsFor_roll3	0.007	0.008	0.023
HomeTeam_ShotsAgainst_roll3	0.034	0.033	0.025
HomeTeam_ShotsFor_roll3	0.002	0.031	0.025

Table 10: Feature Importances for Logistic Regression, Random Forest, and XGBoost continues

Feature	Logistic Regression	Random Forest	XGBoost
HomeTeam_ShotsOnTargetAgainst_roll3	0.000	0.027	0.026
HomeTeam_ShotsOnTargetFor_roll3	0.029	0.025	0.022
HomeTeam_YellowCardsAgainst_roll3	0.031	0.022	0.022
HomeTeam_YellowCardsFor_roll3	0.007	0.022	0.021
HomeTeam_enc	0.009	0.024	0.025
Home_pre-po	0.087	0.033	0.029
Month	0.065	0.021	0.027
Overall Rating_Away	0.163	0.033	0.039
Overall Rating_Home	0.166	0.032	0.041
matchday	0.038	0.031	0.021

Table 11: Top-5 Odds Differences (VC – RF) by Market

Market	Date	Fixture	RF_Odds	VC_Odds	Δ (VC–RF)
Home Win	2023-12-23	Eupen vs. St. Gilloise	3.33	12.00	+8.67
Home Win	2023-12-26	Kortrijk vs. Gent	4.25	8.50	+4.25
Home Win	2024-01-28	Eupen vs. Antwerp	3.92	8.00	+4.08
Home Win	2023-12-10	Eupen vs. Genk	3.55	7.50	+3.95
Home Win	2023-10-21	Kortrijk vs. Club Brugge	3.86	7.50	+3.64
Draw	2024-01-30	Club Brugge vs. Kortrijk	5.30	9.00	+3.70
Draw	2023-10-20	St. Gilloise vs. Eupen	3.94	6.50	+2.56
Draw	2024-02-10	Club Brugge vs. Eupen	3.69	6.00	+2.31
Draw	2023-08-05	Genk vs. Eupen	3.23	5.50	+2.27
Draw	2024-03-16	Anderlecht vs. Kortrijk	3.80	6.00	+2.20
Away Win	2024-01-30	Club Brugge vs. Kortrijk	8.91	20.00	+11.09
Away Win	2024-01-31	St. Gilloise vs. RWD Molenbeek	5.63	15.00	+9.37
Away Win	2023-12-17	Genk vs. Kortrijk	4.58	12.00	+7.42
Away Win	2023-10-20	St. Gilloise vs. Eupen	8.40	15.00	+6.60
Away Win	2024-02-10	Club Brugge vs. Eupen	6.54	13.00	+6.46

Codes and Data

Please click on this [Github link](#) to veiw the codes, Data, and other results stated in this report.